HW 1 — Faraday Rotation & Jackson 7.3

Problem 1. Rotation rate in the Faraday example

> In the Faraday-rotation example, the medium supports left-circular polarization with index $\mathbf{n_L}$ and right-circular with $\mathbf{n_R}$. The wave enters at z = 0 linearly polarized along $\hat{x}$. With $\tan\theta = E_y/E_x$ (both real), find θ(z) — i.e. how fast the plane of polarization turns with z.

Given (from lecture):

$$\vec{E}(z,t) = \frac{\sqrt{2}}{2} E \hat{e}_+ e^{i(k_L z - \omega t)} + \frac{\sqrt{2}}{2} E \hat{e}_- e^{i(k_R z - \omega t)}$$

$$\hat{e}_\pm = \frac{1}{\sqrt{2}} (\hat{x} \pm i\hat{y}), \quad k_{L,R} = \frac{\omega n_{L,R}}{c}$$

Step 1 — take Cartesian components.

$$E_x(z,t) = \frac{E}{2} \left(e^{i k_L z} + e^{i k_R z} \right) e^{-i\omega t}$$

$$E_y(z,t) = i \frac{E}{2} \left(e^{i k_L z} - e^{i k_R z} \right) e^{-i\omega t}$$

Step 2 — factor out the mean phase. Let

$$\bar{k} \equiv \frac{k_L + k_R}{2}, \quad \Delta k \equiv k_L - k_R \quad \text{so that} \quad k_{L,R} = \bar{k} \pm \frac{\Delta k}{2}$$

$$e^{i k_L z} + e^{i k_R z} = 2 e^{i \bar{k} z} \cos\frac{\Delta k z}{2}$$

$$e^{i k_L z} - e^{i k_R z} = 2i e^{i \bar{k} z} \sin\frac{\Delta k z}{2}$$

$$\Rightarrow \quad E_x = E e^{i(\bar{k} z - \omega t)} \cos\frac{\Delta k z}{2}, \quad E_y = -E e^{i(\bar{k} z - \omega t)} \sin\frac{\Delta k z}{2}$$

$$\vec{E}(z,t) = E\, e^{i(\bar{k}z-\omega t)} \left[\hat{x} \cos\frac{\Delta k z}{2} - \hat{y} \sin\frac{\Delta k z}{2} \right]$$

Both components carry the same complex factor $E\, e^{i(\bar{k}z-\omega t)}$, so once it is pulled out E_x and E_y are real — the wave stays **linearly polarized at every z**, only its direction turns. That is exactly the condition stated in the problem.

Step 3 — the rotation angle.

$$\tan\theta(z) = \frac{E_y}{E_x} = -\tan\frac{\Delta k z}{2} = \tan\left(-\frac{\Delta k z}{2}\right)$$

$$\theta(z) = -\frac{(k_L - k_R)\, z}{2} = \frac{\omega}{2c}\,(n_R - n_L)\, z$$

Rotation "speed" in z — θ is *linear* in z, so the rate is constant:

$$\frac{d\theta}{dz} = -\frac{k_L - k_R}{2} = -\frac{\omega}{2c}\,(n_L - n_R) \qquad \text{rad/m}$$

$$\left|\, d\theta/dz \,\right| = \frac{\omega}{2c}\left|\, n_L - n_R \,\right| = \frac{\pi}{\lambda_0}\left|\, n_L - n_R \,\right|$$

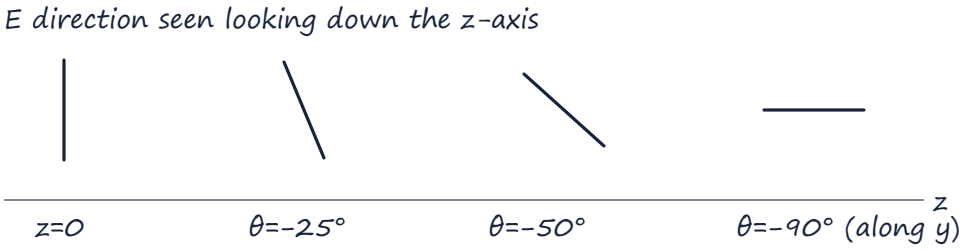
Over a cell of length d the plane of polarization turns by $\beta = \omega(n_R - n_L)d/2c$ — the angle marked β on the "Application" slide. In a magnetized medium $n_L - n_R \propto B$, which gives the Verdet-constant form $\theta = V B d$.

Check against the lecture result.

The wave is polarized along $\hat{y}$ when $E_x = 0$, i.e. $\cos(\Delta k z/2) = 0$:

$$\frac{(k_L - k_R)\, z}{2} = \frac{\pi}{2} + m\pi \;\Rightarrow\; (k_L - k_R)\, z = (2m+1)\pi \;\Rightarrow\; z = \frac{(2m+1)\pi}{\frac{\omega}{c}(n_L - n_R)}$$

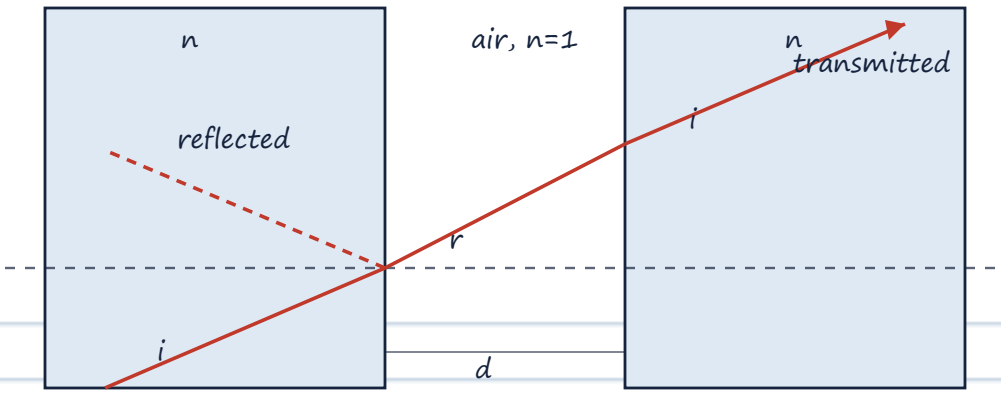
✓ Same as the slide. At those z, $\theta = -(2m+1)\pi/2$, i.e. 90° from $\hat{x}$ — along $\hat{y}$.



The polarization plane turns uniformly with z at the constant rate dθ/dz.

Problem 2. Jackson 7.3 — air gap between two identical slabs

> **7.3** Two plane semi-infinite slabs of the same uniform, isotropic, nonpermeable, lossless dielectric with index of refraction n are parallel and separated by an air gap (n = 1) of width d. A plane wave of frequency ω is incident on the gap from one of the slabs with angle of incidence i. For linear polarization *both* parallel to *and* perpendicular to the plane of incidence,
> **(a)** calculate the ratio of power transmitted into the second slab to the incident power and the ratio of reflected to incident power;
> **(b)** for i greater than the critical angle for total internal reflection, sketch the ratio of transmitted power to incident power as a function of d measured in units of wavelength in the gap.



Step 1 — geometry (Snell at both faces).

$n \sin i = 1 \cdot \sin r \;\Rightarrow\; \sin r = n \sin i, \quad \cos r = \sqrt{1 - n^2 \sin^2 i}$

at the 2nd face: $1 \cdot \sin r = n \sin i' \;\Rightarrow\; i' = i$ (exit ∥ entry, media 1 & 3 identical)

critical angle: $\sin i_o = 1/n$

Step 2 — single-interface Fresnel coefficients (slab → gap). Using the impedance form from lecture with $Z = Z_o/n$ and $Z' = Z_o$:

$$r_s = \frac{Z' \cos i - Z \cos r}{Z' \cos i + Z \cos r} = \frac{n \cos i - \cos r}{n \cos i + \cos r}$$ (⊥ to plane of incidence)

$$r_p = \frac{Z \cos i - Z' \cos r}{Z \cos i + Z' \cos r} = \frac{\cos i - n \cos r}{\cos i + n \cos r}$$ (∥ to plane of incidence)

Stokes relations (gap → slab is the same interface reversed — and slab 3 = slab 1):

$r_{21} = - r_{12}\,, \quad t_{12}\, t_{21} = 1 - r_{12}^2$ (true for both s and p)

Step 3 — sum the multiply-reflected waves inside the gap.

One crossing of the gap multiplies the amplitude by $e^{i\beta}$, where β is the phase picked up by the z-component of $\vec{k}$ in the gap:

$$\beta = k_{gap,z}\, d = \frac{\omega d}{c}\cos r = \frac{2\pi d}{\lambda_o}\cos r$$

$$t_{tot} = t_{12}t_{21}e^{i\beta}\sum_{m=0}^{\infty}\left(r_{21}^{\,2}e^{2i\beta}\right)^m = \frac{(1-r^2)\,e^{i\beta}}{1-r^2e^{2i\beta}}$$

$$r_{tot} = r + \frac{t_{12}t_{21}r_{21}e^{2i\beta}}{1-r^2e^{2i\beta}} = \frac{r\,(1-e^{2i\beta})}{1-r^2e^{2i\beta}}$$

Because slab 3 is the same medium as slab 1 and the exit angle equals i, the flux factor $(Z/Z')(\cos r/\cos i)$ in the transmission coefficient is exactly 1, so $T = |t_{tot}|^2$ and $R = |r_{tot}|^2$.

Step 4 — take the modulus ($i < i_o$, so r and β are real):

$$\left|\,1 - r^2e^{2i\beta}\,\right|^2 = 1 - 2r^2\cos 2\beta + r^4 = (1-r^2)^2 + 4r^2\sin^2\beta$$

$$T = \frac{1}{1 + F\sin^2\beta}\,, \qquad R = \frac{F\sin^2\beta}{1 + F\sin^2\beta} = 1 - T\,, \qquad F \equiv \frac{4r^2}{(1-r^2)^2}$$

Step 5 — evaluate F for each polarization. With $1 - r_s^2 = 4n\cos i\cos r/(n\cos i + \cos r)^2$:

$$F_s = \frac{(\,n^2\cos^2 i - \cos^2 r\,)^2}{4n^2\cos^2 i\cos^2 r}\,, \qquad F_p = \frac{(\,\cos^2 i - n^2\cos^2 r\,)^2}{4n^2\cos^2 i\cos^2 r}$$

Now use $\cos^2 r = 1 - n^2\sin^2 i$. The numerators collapse beautifully:

$$n^2\cos^2 i - \cos^2 r = n^2\cos^2 i - 1 + n^2\sin^2 i = n^2 - 1$$

$$\cos^2 i - n^2\cos^2 r = (n^2-1)\left[\,(n^2+1)\sin^2 i - 1\,\right]$$

$$\text{(a)}\quad F_s = \frac{(n^2-1)^2}{4n^2\cos^2 i\cos^2 r}\,, \qquad F_p = \frac{(n^2-1)^2\left[(n^2+1)\sin^2 i - 1\right]^2}{4n^2\cos^2 i\cos^2 r}$$

$$T_{s,p} = \left[\,1 + F_{s,p}\sin^2\beta\,\right]^{-1}, \quad R_{s,p} = 1 - T_{s,p}, \quad \beta = (\omega d/c)\cos r$$

Sanity checks

✓ $d = 0 \Rightarrow \beta = 0 \Rightarrow T = 1$: the slabs touch and there is no interface at all.

✓ $\beta = m\pi$, i.e. $d\cos r = m\lambda_o/2$ (half-wave gap) $\Rightarrow T = 1$ for any n. Fabry–Pérot resonance.

✓ $F_p = 0$ when $(n^2+1)\sin^2 i = 1$, i.e. $\tan i = 1/n$ — **Brewster's angle**. Then $T_p = 1$ for every d, as it must be.

✓ $n \to 1 \Rightarrow F \to 0 \Rightarrow T \to 1$. And at $i = 0$ both reduce to $F = (n^2-1)^2/4n^2$.

7.3 (b) Beyond the critical angle — frustrated total internal reflection

For $i > i_0$ we have $\sin r = n \sin i > 1$, so $\cos r$ becomes pure imaginary (the gap field is the evanescent wave from lecture 4):

$$\cos r = i\gamma , \quad \gamma \equiv \sqrt{n^2 \sin^2 i - 1} \quad \Rightarrow \quad \beta = i\alpha , \quad \alpha \equiv \frac{\omega d \gamma}{c} = 2\pi\gamma \frac{d}{\lambda_0}$$

$$\sin^2\beta = \sin^2(i\alpha) = -\sinh^2\alpha , \quad \cos^2 r = -\gamma^2$$

Everything analytically continues; the two minus signs cancel in $F \sin^2\beta$, so T stays real and positive:

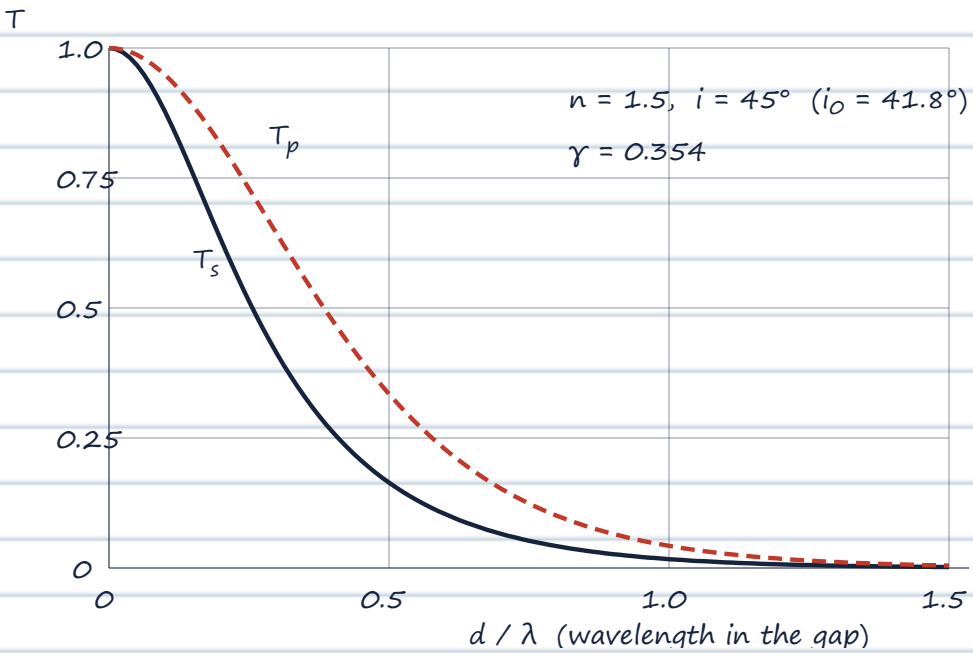
(b)

$$T_s = \left[1 + \frac{(n^2-1)^2}{4n^2\gamma^2\cos^2 i} \sinh^2\alpha \right]^{-1}$$

$$T_p = \left[1 + \frac{(n^2-1)^2[(n^2+1)\sin^2 i - 1]^2}{4n^2\gamma^2\cos^2 i} \sinh^2\alpha \right]^{-1} , \quad \alpha = 2\pi\gamma\, d/\lambda$$

Since the gap is vacuum, the wavelength in the gap is λ_0, so d/λ is exactly the abscissa Jackson asks for. **No oscillation** — sinh is monotonic, so T falls smoothly from 1 to 0.

$d \gg \lambda$: $\sinh^2\alpha \to \tfrac{1}{4}e^{2\alpha} \Rightarrow T \approx \dfrac{16\, n^2\gamma^2\cos^2 i}{(n^2-1)^2}\, e^{-4\pi\gamma d/\lambda}$ photon tunnelling



(b) T vs. gap width. Monotonic exponential decay — T_p always exceeds T_s.

Representative values for the sketch ($n = 1.5$, $i = 45°$): $d/\lambda = 0.1 \to T_s = 0.88$, $T_p = 0.95 \cdot 0.25 \to 0.51$, $0.73 \cdot 0.5 \to 0.16$, $0.33 \cdot 1.0 \to 0.017, 0.043$.